

pyharm: Analysis and Plotting for GRMHD

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Summary

pyharm is a scalable python library for plotting and analyzing some general-relativistic magnetohydrodynamic (GRMHD) simulations, especially those simulating matter around black holes. It handles reading a number of different simulation snapshot formats, plotting basic and derived fluid properties in different spacetimes and coordinate systems, and computing arbitrary reductions across large datasets of fluid snapshots.

Statement of Need

As large, complex 3D datasets, GRMHD simulation snapshots require sophisticated post-processing and plotting code in order to correctly interpret their data values. Writing new plotting scripts from scratch takes time and effort, not to mention being bug-prone due to the overall complexity. A framework which pre-defines variables and coordinates consistently saves time in understanding these datasets, and improves the quality of results by allowing fixes to be preserved in all future calculations.

State of the Field

Many plotting libraries and utilities exist for visualizing and analyzing eulerian hydrodynamics simulations at scale, e.g. ([Turk et al., 2010](#)), ([Bethel et al., 2012](#)), ([Ahrens et al., 2005](#)). However, GRMHD in particular presents two unique challenges that make these tools difficult to use or wholly insufficient:

1. An analytic transformation is often made from a “natural” coordinate system (r, θ, ϕ in the spacetime) into a separate coordinate system where the simulation grid is regular. Imaging simulations requires performing the reverse transformation.
2. Fluid quantities in GRMHD do not generally match the quantities in Newtonian MHD, and generally require the local metric information, and potentially frame transformations, to compute.

Plotting scripts exist which natively support GRMHD grids and quantities, but are generally either private, or specialized to a particular code or style of GRMHD ([Bozzola, 2021](#)). pyharm attempts to support as many codes as possible.

Code Design

The central class in pyharm is the FluidState, representing a snapshot of a GRMHD simulation at some point in time (either generated, memory-backed, or file-backed, generally the latter). Data retrieval and manipulation are both done by means of the `__getitem__` function, allowing users to treat the object as a dictionary containing any desired derived variable. When a key is retrieved, e.g. `dump['beta']` for the plasma β parameter (ratio of gas pressure to

37 magnetic pressure), it is computed on the fly, returned, and cached in memory for the next
38 use. Quantities can be defined in terms of one another recursively, and thus the code avoids
39 repeating common calculations.

40 While pyharm is designed first as a python library for user scripts, it has gained a number of
41 useful command-line utilities, two of which are easily extensible and broadly useful as a way of
42 accessing pyharm's functionality.

43 pyharm movie handles slice and average plots per-snapshot across a simulation (including
44 optionally encoding the results using ffmpeg). Large jobs consisting of many frames can be
45 parallelized across multiple cores or nodes with Python's built-in multiprocessing or with
46 mpi4py.futures (Rogowski et al., 2023). New movies can be defined solely by adding a
47 function in pyharm/plots/figures.py which creates one frame, with the parallelism features
48 shared between all functions.

49 pyharm analysis handles reductions over snapshots from an entire simulation (slices, time-
50 averages, flow rates, etc.). Results are written to a single HDF5 file collecting from all
51 processes/ranks, in a standard layout readable by another command-line script, pyharm plot-
52 result.

53 In addition to the main scripts, pyharm provides a number of small utilities for checking
54 simulation logs for progress and errors, checking validity of simulation snapshots, or checking
55 snapshots against one another.

56 Implementation Notes

57 pyharm is written in pure Python using numpy, scipy, and matplotlib. As it is implemented
58 purely with array operations, not loops, the code shares numpy and scipy's good efficiency
59 on single CPU cores. Parallelizing operations on a single dump is not a goal of the code –
60 usually, compute-intensive operations are conducted across many dump files in a way which
61 is substantially or wholly embarrassingly parallel. Likewise, GPUs are not a priority as many
62 operations are already significantly bottlenecked by file read speed. Instead, pyharm achieves
63 good speed mostly by avoiding unnecessary read and compute operations.

64 As its most important optimization, pyharm supports slicing of HDF5-based dump files without
65 first reading the entire grid into memory. Thus if only, e.g., a slice at $\phi = 0$ is required, pyharm
66 supports a slicing operation `sliced_dump = dump[:, :, 0]` before any reads are performed, and
67 will read only the given slice from the data file. Most operations are supported on sliced files,
68 including the calculation of derived variables, reductions, and plotting.

69 In addition, the (sliced) arrays corresponding to each key are natively cached by the pyharm
70 data objects: file readers can cache the variables retrieved by a FluidState object, and
71 FluidState objects themselves can cache the results of computations to avoid repeating them.
72 Caching can be memory-intensive, but when combined with slicing it can lead to dramatic
73 improvements in runtime of complex analyses and reductions which share basic components.

74 With these optimizations, a basic variable from a low-resolution (192x96x96 zones) run of
75 2000 snapshot files can be plotted in as little as a minute on a 12-core desktop computer, and
76 a single node of Harvard Cannon (56 cores) images a more substantial simulation (192x96x96
77 zones, ~3000 snapshots) in about a minute as well – in each case, more than enough to
78 generate a 30fps movie as it plays. A much more complex analysis (about 70 reductions over
79 derived variables) completes in 20 minutes on the desktop and 31 minutes on the cluster.

80 Tests

81 As primarily a plotting framework, most bugs or undesired behaviors in pyharm cannot be
82 easily tested in an automated way. However, calculations of physical quantities are tested

83 against expectations, such as an example calculation of 4-vector quantities with known results.
84 “Golden file” regression tests of the calculation of basic reductions are also available, over an
85 array of different file formats.

86 Research projects using pyharm

87 pyharm has been used in obtaining the results of many published papers, especially from the
88 GRMHD code KHARMA (Prather, 2025). This includes collaboration papers from the EHT
89 such as (Event Horizon Telescope Collaboration et al., 2019) and (Event Horizon Telescope
90 Collaboration et al., 2022), and independent papers using KHARMA, e.g. most recently
91 (Thomas et al., 2026), (Bukowiecka et al., 2026) and (Stanway et al., 2026).

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109 AI usage disclosure

110 No AI tools were used in developing pyharm or this paper.

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